Fiscal Year:	FY 2007	**Task Last Updated:**	FY 03/06/2008
PI Name:	Dutta, Prabir Ph.D.		
Project Title:	Bioinspired Zeolite Assemblies as Artificial Photosynthetic Systems		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Bio-nanotechnology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	dutta.1@osu.edu	**Fax:**	FY 614-688-5402
PI Organization Type:	UNIVERSITY	**Phone:**	614-292-4532
Organization Name:	The Ohio State University		
PI Address 1:	Department of Chemistry		
PI Address 2:	100 West 18th Avenue		
PI Web Page:	http://www.chemistry.ohio-state.edu/		
City:	Columbus	**State:**	OH
Zip Code:	43210-1173	**Congressional District:**	15
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-UNSOL
Start Date:	01/01/2004	**End Date:**	09/30/2007
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Gilles, Donald	**Contact Phone:**	256 544-9302
Contact Email:	donald.C.Gillies@msfc.nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2007 per PI (11/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	George, Michael (University of Alabama in Huntsville)		
Grant/Contract No.:	(MSFC) NAG8-1940 (JSC) NNJ04HG97G		
Performance Goal No.:			
Performance Goal Text:			
	Our goal is to assemble a supramolecular structure around a central zeolite membrane with the following features. · Energy transfer from extrazeolitic molecules to sensitizer molecules on the surface of the zeolite membrane · Charge (electron) transfer from the photoexcited sensitizer molecule to acceptor molecules within the zeolite membrane · Charge transport through the membrane by electron hopping · Catalytic oxidation and reduction of H2O to oxygen and hydrogen on two opposite sides of the membrane and regeneration of the donor and acceptor		

Task Description:

· Zeolite pore engineering to ensure entrapment of photoactive molecules, but still allow for proton migration to maintain charge balance

The task is divided into four subcategories, dealing with zeolitic energy transfer, electron transfer processes on the surface of the zeolite, intrazeolitic charge propagation and zeolite-based nanostructured catalysts. Choice of molecules include polypyridyl complexes of Ru and Os as sensitizers and electron donors and bipyridinium-based systems as electron acceptors and metal oxides as catalysts.

Photosynthesis, the conversion of sunlight to useful chemical energy is the basis for life. Important concepts that have emerged from studies on natural photosynthesis include the structure of the photoactive molecules, novel spatial arrangement, directed electron transfer and catalysis. Our objective is to design an artificial photosynthetic assembly with the goal of photolytic splitting of water into hydrogen and oxygen. We propose to use microporous crystalline, aluminosilicate zeolites as the host system. The challenge in design of complex, multicomponent functional assemblies from nanoscale building blocks is the optimization and assembly of the various components. Since the ordered array of channels and cages exist in nanocrystallites as well as in larger zeolite membranes, a convenient template for self assembly over large length scales becomes possible. Choice of molecules in this study include polypyridyl complexes of Ru and Os as sensitizers and electron donors and bipyridinium-based systems as electron acceptors. The nanometer size cages and channels of aluminosilicate zeolites will anchor the photochemical moieties as well as the catalysts necessary for reaction with charge separated redox species. The specific objectives of the research program includes:

$ energy transfer within nanocrystalline (~20-100nm) zeolite crystallites, as well as energy transfer into and out of a zeolite

$ electron transfer dynamics from surface-bound sensitizers into acceptor molecules in zeolites

$ synthesis of zeolitic membranes from nanocrystalline zeolites and photochemical charge propagation within zeolite membranes

$ use of nanometer sized zeolite cages for synthesis of metal oxide catalysts for multi-electron processes

· assembly of the above components into a multifunctional unit capable of light-driven conversion of water to H2 and O2.

Long term space exploration and habitation necessitate that systems be developed for generation of molecular building blocks that can be assembled to form complex macroscopic structures, such as fuel, polymers, drugs, etc. Ideally, the process should begin with the simplest of molecules that are readily available and use a plentiful energy source. The best example in nature of such a process is photosynthesis, which, in plants, involves the solar energy-driven catalytic reaction of making carbohydrates from water and carbon dioxide. This is the basis on which life is sustained on earth. The conversion of H2O and CO2 to O2 and carbohydrates is a complex redox process, involving a large array of molecules and a complex membrane-based architecture. In principle, many solar energy-based transformations are possible, in which simple molecules like H2O, CO2, N2 can be converted to molecular building blocks such as H2, CH4, NH3, which can form the basis for manufacture of important chemicals necessary for life.

Solar radiation can drive these uphill reactions. The simplicity of these reactions masks the complexity of the architecture necessary to accomplish the goal. However, natural photosynthesis provides significant clues for assembly of "artificial" photosynthetic systems, and in this proposal, we propose a zeolite-based architecture, that at a primitive level mimics natural photosynthesis. For our model redox reactions, we have chosen the splitting of H2O into H2 and O2. The principles we develop in this program should be applicable to other photoredox systems, especially the architectural aspects related to assembly of multifunctioning, multicomponent units. The limitations of weight that can be carried on long term missions as well as space habitation require that materials taken from earth can serve several functions. Zeolites, because of their applications in adsorption, ion-exchange, separations and catalysis can play that role. In addition, photosensitized reactions for both environmental pollution control (air and water purification) as well as synthesis of fine chemicals have been reported with zeolites, and is currently satisfying aspects of “sustainable/green chemistry” on earth .

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Successful artificial photosynthesis systems that can manufacture fuels and chemicals using sunlight and simple chemicals such as water and carbon dioxide as feed will revolutionize our energy dependence. These technologies are examples of "disruptive technologies", since they set up a new paradigm of energy utilization.

Project I. An Integrated Zeolite Membrane - RuO2 Photocatalyst System for Hydrogen Production from Water
Zeolite Mediated Photochemical Charge Separation Using A Surface Entrapped Ruthenium-Polypyridyl Complex

Employing the strategy of quaternization of the 2,2´ N atoms of the conjugated bipyridine ligand, 1,4-bis[2-(4'-methyl-2,2'-bipyrid-4-yl)ethenyl]benzene (L), a polypyridyl complex of ruthenium (II) has been tethered on the surface of zeolite Y. Electrochemical and spectroscopic properties of the ligand suggest that upon visible photoexcitation of the MLCT band, the electron is localized on the conjugated ligand rather than the bipyridines. Electron transfer from the surface complex to bipyridinium ions (methylviologen) within the zeolite was observed. Visible light photolysis of the ruthenium-zeolite solid ion-exchanged with diquat and suspended in a propylviologen sulfonate solution led to permanent formation of the blue propylviologen sulfonate radical ion in solution. The model that is proposed involves intrazeolitic charge transfer to ion-exchanged diquat followed by interfacial (zeolite to solution) electron transfer to propylviologen sulfonate in solution. Because of the slow intramolecular back electron transfer reaction and the forward electron propagation via the ion-exchanged diquat, Ru(III) is formed. This Ru(III) complex formed on the zeolite is proposed to react rapidly with water in the presence of light, followed by reaction with the propylviologen sulfonate to form pyridones and regeneration of Ru(II), which then continues the photochemical process.

Photochemical Studies with a Zeolite Y Membrane Formed via Secondary Growth

Task Progress:	Zeolite membranes offer an attractive feature in design of artificial photosynthetic assemblies because they can keep the charge separated species apart, yet allow for communication between the two sides via the intervening zeolite channels and cages. However, such a strategy would require that the zeolite membrane be pinhole-free. In an earlier paper (J. Phys. Chem. B, 106, 11898, 2002), we have shown that nanocrystalline zeolite Y can be spin-coated on porous substrates to form membranes, and the pinholes can be plugged by photoresist. Such membranes do not allow manipulation at higher temperatures or even in solution chemistry because of the thermal instability and solubility characteristics of the photoresist. In this paper, we report on a zeolite Y membrane made by the secondary growth method. The growth process leads to minimal pinholes as determined by leak tests. The sturdiness of the membranes allows for manipulation and we have shown that organic reactions can be carried out in the zeolite pores. Charge transport through the membrane by an electron hopping process mediated via self-exchange of ion-exchanged bipyridinium ions are demonstrated. The electron transfer was initiated using a photochemical Ru(bpy)32+ - EDTA system. Zeolite Membrane Based Artificial Photosynthetic Assembly For Long-Lived Charge Separation Photochemically generated long-lived charge separation is the key step in processes that aim for conversion of solar into chemical energy. In this study, we focus on a Ru polypyridyl complex [(bpy)2 Ru(II)L , bpy = bipyridine, L = 1, 2-bis [4-(4' – 2,2' – bipyridyl) ethene] encapsulated on the surface of a pin-hole free zeolite membrane by quaternization of L and surrounded with intrazeolitic bipyridinium ions (N,N'-trimethyl-2,2'-bipyridinium ion, 3-DQ2+). Visible-light irradiation of the Ru complex side of the membrane in the presence of a sacrificial electron donor led to formation of PVS? • on the other side. Pore-blocking disilazane-based chemistry allows for Na+ to migrate through the membrane to maintain charge balance, while keeping the 3-DQ2+ entrapped in the zeolite. These results provide encouragement that the zeolite-membrane based architecture has the necessary features for not only incorporating molecular assemblies with long-lived charge separation, but also for ready exploitation of the spatially separated charges to store visible light energy in chemical species. Controlled Release of Paraquat from Surface-Modified Zeolite Y The surface hydroxyl groups of zeolite Y were reacted with 1,1,3,3-tetramethyldisilazane, HN(SiHMe2)2 under ambient conditions and the grafting of siloxy functionality on the zeolite was confirmed by infrared, NMR spectroscopy and elemental analysis. Paraquat (methyl viologen) was ion-exchanged into the zeolite, followed by treatment with the disilazane reagent. Surface modification of paraquat-loaded zeolites encapsulates the guest molecules in the zeolite cages and release of paraquat by ion-exchange with sodium ions was studied. The amount of paraquat released was dependent on the concentration of Na+ in solution, and was similar for the derivatized and underivatized samples. In the absence of surface modification, equilibration occurred within minutes, whereas with surface modification, the equilibration time was extended to a week. Paraquat was chosen as the guest molecule, since it is widely used as an herbicide and its controlled release is of interest in agricultural applications. An Integrated Zeolite Membrane - RuO2 Photocatalyst System for Hydrogen Production from Water An integrated zeolite membrane on a RuO2 containing alumina support is described. Upon visible light illumination of Ru(bpy)32+, N, N1-trimethyl-2, 21-bipyridinium and EDTA on the zeolite side of the membrane, charge is propagated through the zeolite membrane and transferred to propylviologen sulfonate within the pores of the alumina support. Nanofingers of ruthernium oxide is assembled within the alumina support by thermal decomposition of ruthenium dodecacarbonyl followed by oxidation of the metal. Hydrogen evolution was monitored on the alumina side as visible light illumination is incident on the zeolite side of the membrane. Project II. Visible-light-Driven Photoreactions of [(bpy)2Ru(II)L]Cl2 in aqueous solutions (bpy = bipyridine, L = 1,2-bis(4-(4'-methyl)-2,2'-bipyridyl) ethene) Visible-light-induced photoreactions of [(bpy)2Ru(II)L]Cl2 (bpy = bipyridine, L = trans-1,2-bis(4-(4'-methyl)-2,2'-bipyridyl) ethene) in aqueous solutions is examined. Reactions are monitored by absorption and emission spectra at different pH; Photolysis of Ru complex in solutions at pH 7 and 12, leads to increased emission with ~50 nm blue-shift of the emission maximum to 620 nm and a less intense absorption band at 330 nm the latter indicating disruption of the olefinic C=C bond of L ligand. Spectral changes occur more rapidly at higher pH, whereas no changes are observed in solutions at pH < 4. From fluorescence titrations, it is found that Ru complex is stronger base (pKa* = 7) in the excited state than in the ground state (pKa = 4). With the help of HPLC, MS, Raman and NMR, photoproducts have been analyzed and it is found that photoinduced isomerization, dimerization are involved. Such reactions do not occur in dark or in aprotic solvent, e.g acetonitrile. We propose that a radical intermediate is formed by photoinduced excited-state electron transfer and proton transfer, which not only accounts for the formation of various photoproducts, and the effects of solvent, pH and light. Significant changes of absorption and emission spectral characteristics of [(bpy)2RuL]2+ in aqueous solutions under visible-light irradiation was observed. The photoreactions exhibited a pH dependence with no reaction below pH 4 and increasing spectroscopic changes under more basic conditions. Using spectroscopic, chromatographic and mass spectrometric methods, products from photoreaction at neutral pH were isolated, with the major product identified as the dimer of [(bpy)2RuL]2+. The mechanism proposed involves a radical intermediate formed via protonation in the excited state and electron localization on the double bond, and explains the observed solvent, pH effect, the spectroscopic data and pathways to observed products. Below pH 4, no photodegradation of [(bpy)2RuL]2+ was observed and emission quenching occurred by an internal electron transfer to a viologen-type moiety formed upon excited state protonation.
Bibliography Type:	Description: (Last Updated: 03/10/2008)
Articles in Peer-reviewed Journals	Kim Y, Lee H, Dutta PK, Das A. "Zeolite-mediated photochemical charge separation using a surface-entrapped ruthenium-polypyridyl complex." Inorg Chem. 2003 Jun 30;42(13):4215-22. PMID: 12817983 , Jun-2003
Articles in Peer-reviewed Journals	Dutta PK, Kim Y. "Photochemical processes in zeolites: new developments." Current Opinion in Solid State and Materials Science 2003 Dec;7(6):483-90. http://dx.doi.org/10.1016/j.cossms.2004.02.004 , Dec-2003

Articles in Peer-reviewed Journals	Kim Y, Dutta PK. " Photochemical studies with a zeolite Y membrane formed via secondary growth." Research in Chemical Intermediates, 2004;30(2):147-61. http://dx.doi.org/10.1163/156856704322960736 , Dec-2004
Articles in Peer-reviewed Journals	Kim Y, Das A, Zhang H, Dutta PK. "Zeolite membrane-based artificial photosynthetic assembly for long-lived charge separation." J Phys Chem B. 2005 Apr 21;109(15):6929-32. PMID: 16851782 , Apr-2005
Articles in Peer-reviewed Journals	Zhang H, Kim Y, Dutta PK. "Controlled release of paraquat from surface-modified zeolite Y." Microporous and Mesoporous Materials 2006 Jan 21;88(1-3): 312-8. http://dx.doi.org/10.1016/j.micromeso.2005.09.026 , Jan-2006
Articles in Peer-reviewed Journals	Kim Y, Dutta PK. "An integrated zeolite membrane/RuO2 photocatalyst system for hydrogen production from water. " Journal of Physical Chemistry C 2007 Jul 19;111(28):10575-81. http://dx.doi.org/10.1021/jp0703096 , Jul-2007